

Decision-Making Under Uncertainty: Tactical Intelligence Transfer from Football to Business

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Overview

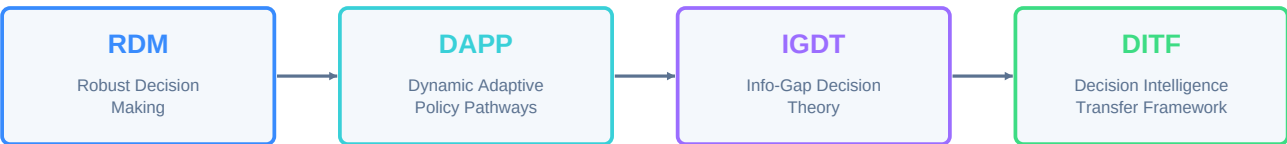
This thesis examines how decision-making under uncertainty in football can inform strategic and operational decision-making in business. Both contexts involve rapid judgment under pressure, limited information, and constantly changing conditions. By studying twenty competitive matches between 2014 and 2023, this research builds a framework that connects tactical intelligence in sport with managerial decision-making in business.

Critical turning points (goals, substitutions, red cards, tactical changes) are treated as **decision events** that shape outcomes. These events are interpreted qualitatively using structured data from Wyscout reports, then mapped to equivalent business situations such as crisis response, leadership under pressure, or recovery after setbacks. Five expert interviews validate these mappings for practical relevance.

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High-stakes matches analyzed (2014-2023)	Expert interviews for cross-domain validation	Thematic clusters of decision events	DMDU frameworks: RDM · DAPP · IGDT

Methodological Foundation

The methodology draws on **Decision-Making under Deep Uncertainty (DMDU)** approaches — Robust Decision Making (RDM), Dynamic Adaptive Policy Pathways (DAPP), and Info-Gap Decision Theory (IGDT) — which emphasize flexibility, adaptability, and learning over prediction and control. These three frameworks converge into the thesis's core contribution: the Decision Intelligence Transfer Framework (DITF).



Five Guiding Research Questions

RQ1	How can football match data be structured and modelled to simulate decision-making under uncertainty?
RQ2	What types of football decisions reflect real-world high-stakes business decisions?
RQ3	How can critical turning points be used to analyse decision impact in simulated environments?
RQ4	What methods can build a transferable decision intelligence framework from sport to business?

Case Structure: Four Thematic Clusters

Twenty matches are organized into four clusters, each representing a distinct decision-making context. Each match is treated as a case study of turning points and the responses they triggered.

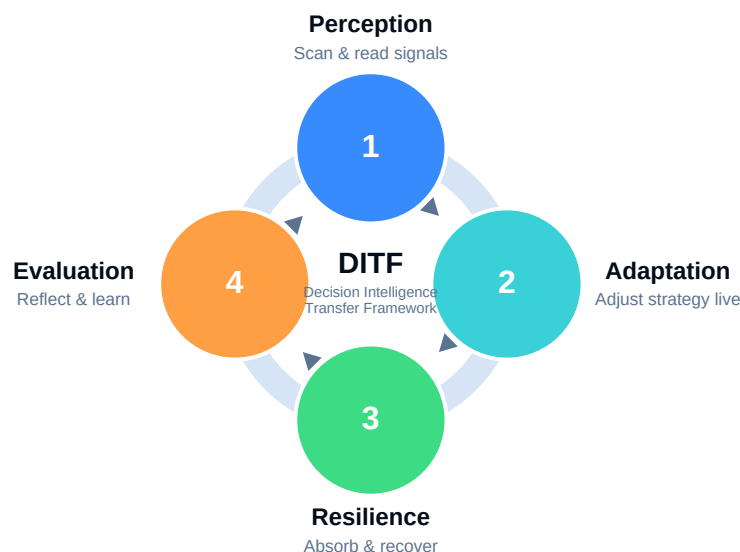


20 high-stakes matches (2014-2023) analyzed across 4 thematic clusters of decision events

Crisis & Recovery	Crisis management emerges as structured improvisation rather than rigid control — high performers operate effectively within uncertainty rather than eliminating it.
Tactical Adaptation	Adaptation appears as structured flexibility: managers reinterpret plans dynamically through structured intuition and time-bound recalibration.
Leadership Under Pressure	Reveals the interaction between emotional steadiness, adaptive structure, and learned resilience — composure scales from individuals to teams.
Controversy & Unpredictability	Shows that ethics is embedded within decision stability — fairness and trust act as stabilizers when procedural clarity fails.

The Decision Intelligence Transfer Framework (DITF)

The DITF is the integrative outcome of this research — a domain-agnostic, cyclical model for navigating complex and volatile environments. It transfers decision mindsets (not surface-level tactics) from sport to business, emphasizing anticipation of uncertainty, adaptive teamwork, and continuous learning over static plans.



1 • Perception	Scanning and interpreting the environment to build situational awareness — reading the game, or monitoring market signals and competitor moves in business.
2 • Adaptation	Modifying tactics or strategy in response to feedback — a substitution on the pitch, or pivoting a project plan when conditions shift in business.
3 • Resilience	Maintaining coherence and recovering quickly after disruption — regaining focus after conceding a goal, or sustaining operations through a crisis.
4 • Evaluation	Assessing outcomes and extracting lessons before the cycle repeats — post-match analysis, or post-project debriefs that refine future decisions.

Key Findings

- **Uncertainty does not diminish decision quality** if it is recognized and structured — teams that remain open to tactical redirection after setbacks recover faster than those anchored to rigid plans.
- **Adaptability is both cognitive and structural** — speed of coordination matters more than speed of action, reinforced through shared understanding and constant communication.
- **Ethics is embedded in resilience** — fairness and legitimacy directly influence perceived stability in uncertain systems, not as a side issue but as a stabilizing structure.
- **Decision intelligence means controlled adaptability, not perfect control** — the DITF adds a behavioural layer to DMDU, combining robust analytical planning with human intuition and moral judgment.

Implications for Business Practice

Structured flexibility	Establish decision protocols with predefined signposts and adaptive pathways (drawn from DAPP) — pivot strategy without abandoning core objectives.
Distributed situational awareness	Recovery from crises depends on shared awareness rather than hierarchical control — empower information flow to shift dynamically toward those closest to critical information.
Ethical resilience as strategy	Predefine ethical thresholds — "adaptation tipping points" — so disruption tests plans without compromising core principles.
Decision rehearsal	Run decision simulations analogous to tactical drills, building cognitive resilience for when traditional planning tools fail.

Implications for Sports Strategy

The DITF reframes football decision-making as structured cognitive processes — perception, evaluation, adaptation, resilience — rather than isolated instinct. Coaches can embed **tactical inflection points** (formation shifts after a red card, substitutions when intensity drops) as adaptive signposts. Data strategy shifts from *prediction* to *preparedness*, preparing strategies that hold across multiple game states rather than optimizing for one. Resilience becomes a trainable tactical skill — explainable AI assists rather than overrides intuition.

Closing Note

This research demonstrates that high-pressure conditions in sport can inform and improve how leaders navigate uncertainty in business. The DITF provides a structured yet flexible bridge between football-based insights and business decision-making — turning uncertainty from a threat into a continuous source of strategic evolution.

Condensed visual summary — full thesis with detailed case analyses, interviews and coding matrices available separately.